One workflow. Four ways to dock.

A 10-15 minute introduction to protein-ligand docking with Anyone Can Dock.

Start with one example

Dock erlotinib into the crystal structure PDB 1M17. ACD prepares the receptor and ligand, runs AutoDock Vina, then lets you inspect predicted poses.

1M17 receptor

erlotinib ligand

pose + score

ONLINE

No install

Fastest for a first try.

LOCAL

Own computer

Same visual workflow; files stay local.

CLI

Terminal

A repeatable command and output folder.

COLAB

Notebook

Follow cells in a browser runtime.

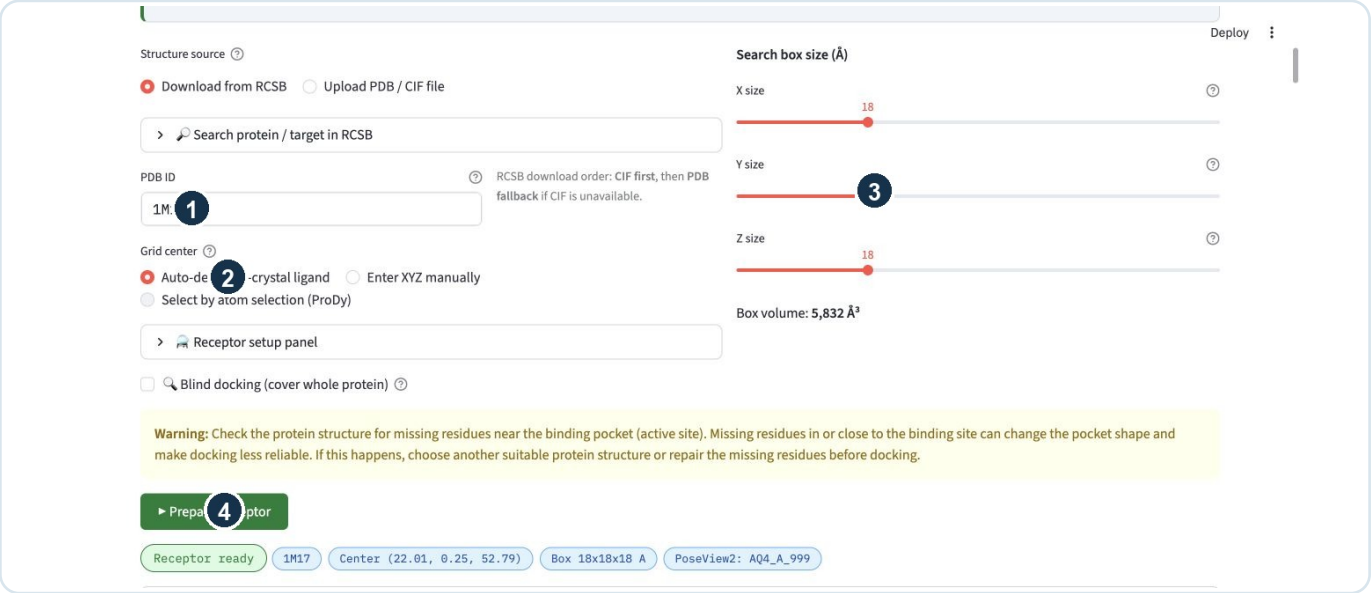
Which should I choose? Start Online. Choose Local for your own machine, CLI for scripts, or Colab for a guided notebook.

Use the same 1M17 + erlotinib example throughout this guide. The four interfaces may show different scores because each docking run samples poses.

ACD Online

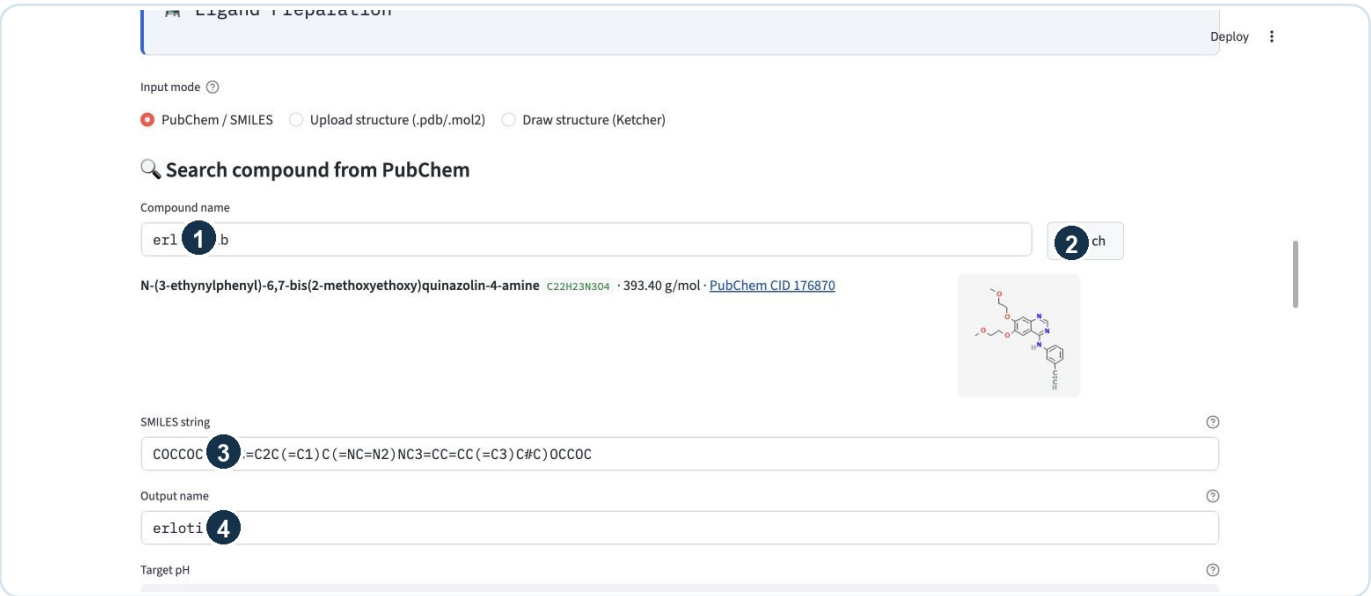
Open the browser app, then follow the numbered controls in these real screenshots.

1 Receptor. In Single Dock, enter 1M17; keep co-crystal centering and the 18 x 18 x 18 Å box; click Prepare Receptor.



1 PDB ID 2 auto center 3 box sizes 4 prepare receptor

2 Ligand. Search erlotinib on PubChem, name the output, keep pH 7.4, then click Prepare Ligand.



1 compound name 2 search 3 returned SMILES 4 output name

3 Run. Keep the basic settings (exhaustiveness 16, 10 poses, energy range 3 kcal/mol). Click Run Docking, then open Results & Visualization to browse poses and downloads.

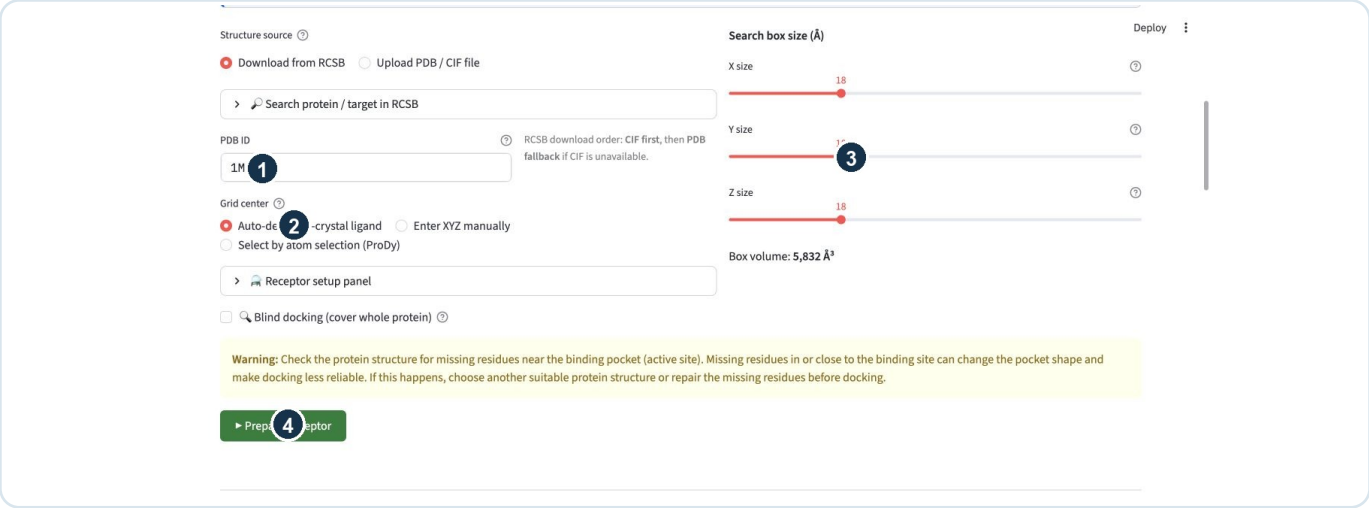
ACD Local

The same interface, running on your own computer.

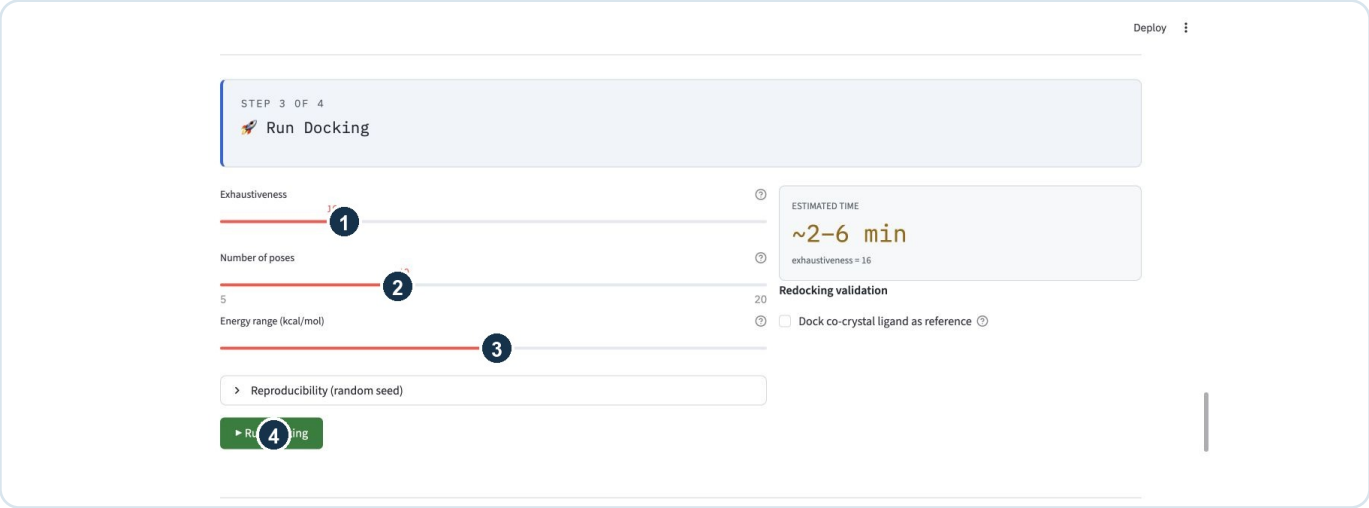
Install Python 3.11, Open Babel, Cairo and Pango first; see the platform-specific setup in the Local repository. Then:

```
git clone https://github.com/nyelidl/anyone-docking-local.git
cd anyone-docking-local
python3.11 -m venv venv
source venv/bin/activate   # macOS / Linux
python -m pip install -r requirements.txt
streamlit run app.py
```

Open the local Streamlit URL shown in Terminal. Use Single Dock and repeat the same 1M17 + erlotinib inputs.



Local screen: PDB ID, auto-centered box, and Prepare Receptor



Local screen: docking settings and the enabled Run Docking button

After the run, inspect the pose viewer and use the result download buttons. Local files are also kept in the app's working directory.

Batch Dock & Co-Dock

Two workflows for more than one ligand.

BATCH DOCK

Dock each ligand separately in the same pocket.
Compare the resulting individual scores.

CO-DOCK

Dock 2-4 ligands together in one pocket.
The score describes the combined pose.

Batch Dock / choose one input

Prepare the receptor, then choose one of the four horizontal input options.

1 SMILES list (text)

Editable SMILES [name] lines; ten examples are prefilled in the 300 px text area.
Example format: CCO ethanol

2 Compound name

Enter one name and click Search. PubChem shows a 2D picture and compound details, then adds it to a separate empty SMILES list.

3 Upload .smi file

Upload a .smi or .txt file with one SMILES [name] per line.

4 Upload structure files

Upload one or more .sdf, .mol2, or .pdb ligand files.

Click Run Batch Docking to score each prepared ligand separately.

Co-Dock / provide 2-4 ligands

Prepare the receptor, then use one of these three input options:

Type names / SMILES

One entry per line. Compound names are looked up in PubChem when you run; SMILES may include a label.

Upload .smi

Upload 2-4 SMILES [name] entries in a .smi or .txt file.

Upload structures

Upload 2-4 .sdf, .mol2, .pdb, or .pdbqt ligand files.

Read the joint result

Run Co-Dock and inspect combined poses.
Joint scores are not per-ligand scores.

ACD CLI

A tested command for a repeatable terminal run.

Install once

```
python -m pip install anyonecandock
```

Run the same practice example

```
acd dock --receptor 1M17 --compound erlotinib \
  --name erlotinib --ph 7.4 --exhaustiveness 4 \
  --poses 5 --yes --output ./erlotinib_dock
```

Read your output folder

erlotinib_scores.csv	Pose scores and ranking.
erlotinib_out.pdbqt	All docked poses in Vina output format.
SDF structures	Ligand structures for viewing or further analysis.

What to check

Open the score table and pose files. A better (more negative) score is a prediction, not experimental binding evidence. Scores and poses can vary between runs.

Prefer a browser?

The next two pages walk through the linked Google Colab notebook with the same 1M17 + erlotinib example.

Google Colab / prepare

Start from the linked notebook, then prepare 1M17 and erlotinib.

0 Set up the notebook

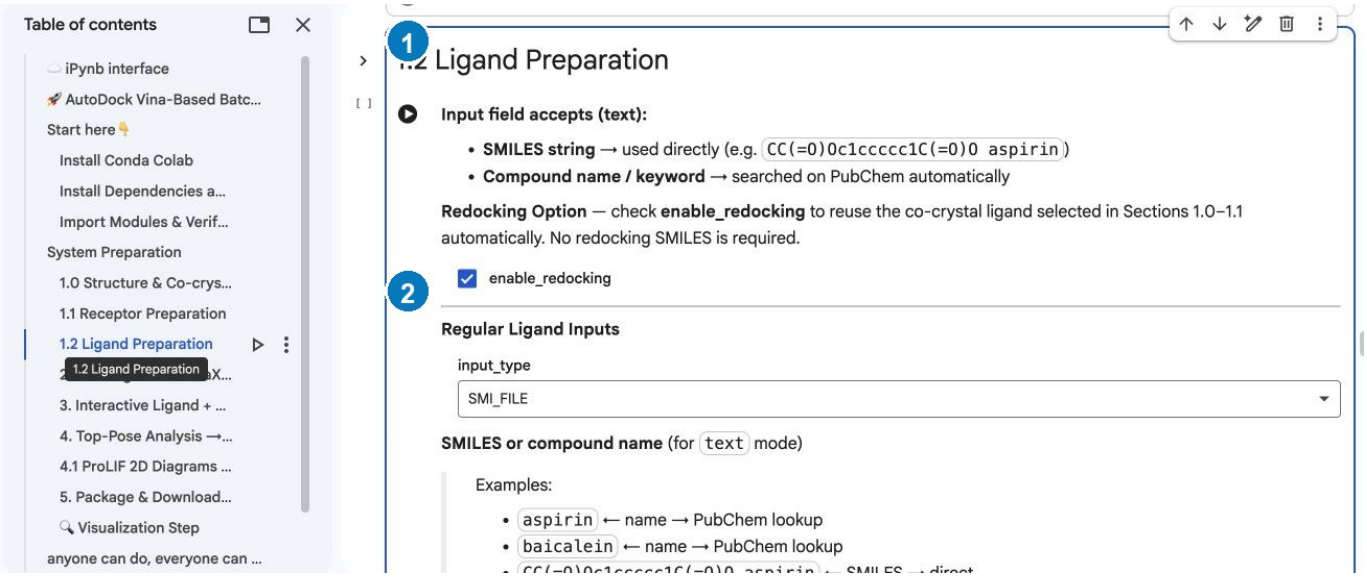
Open the ACD Colab notebook and save a copy in Drive. Connect a runtime. Run Install Conda Colab, Install Dependencies and Docking Engines, then Import Modules & Verify Tools, following any restart prompt.

1 Prepare the receptor

In 1.0 choose Download from RCSB and enter 1M17. In 1.1 keep Yes (use detected ligand) for the box center and 18 x 18 x 18 Angstrom box size. Run both cells.

2 Prepare the ligand

In 1.2 select text input, enter erlotinib, set output_name to erlotinib, and keep target pH 7.4. Uncheck enable_redocking for this simple single-ligand example.



Actual linked Colab view: table of contents and Section 1.2 input controls.

Four engine choices appear in Section 2

VINA, VINAXB, GNINA, and VINA_GPU. Use VINA on a standard runtime for this first run. Vina-GPU is installed on demand and needs a T4 GPU runtime.

Google Colab / dock & read

Choose an engine, inspect poses, and save the complete result package.

3 Run docking

In Section 2 choose VINA, set docking_runs to 1, and uncheck run_redock for this example. Keep exhaustiveness 16 and num_modes 10 for the notebook default. Run the cell.

Available engines

VINA	Standard CPU example; recommended here.
VINAXB	Vina-based alternative scoring.
GNINA	Neural-network scoring option.
VINA_GPU	GPU option; select T4 before setup.

4 Select a pose and interpret results

Run Section 3 to inspect the protein-ligand view and save a selected pose. Section 4 plots top-pose scores; with one run it shows a dot plot, while repeated runs use box plots. Section 4.1 adds ProLIF 2D interactions.

Gemini interpretation

In Section 4, auto_gemini_interpretation can explain the score table when Colab AI is available. Review it against the actual plots and files.

5 Download

Run Section 5 Package & Download Docking Results. Download docking_results.zip from the notebook; it includes staged pose, score, plot, and interaction files. A Gemini Markdown report is included only when generation succeeds.

Docking scores predict poses; they do not establish experimental affinity.

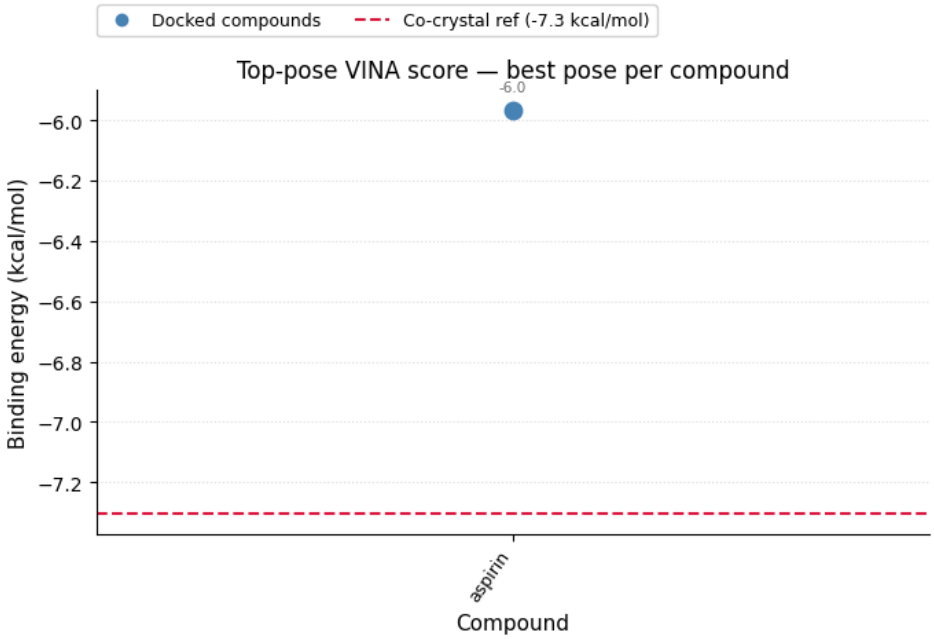
Colab / actual run outputs

Screenshots extracted from the linked notebook after its saved run.

RUN SHOWN HERE

PDB 1M17 + aspirin, VINA, one docking run. Co-crystal ligand AQ4 was also redocked. These are separate from the erlotinib tutorial inputs.

Top-pose score



READ THE PLOT

Aspirin

-5.96 kcal/mol

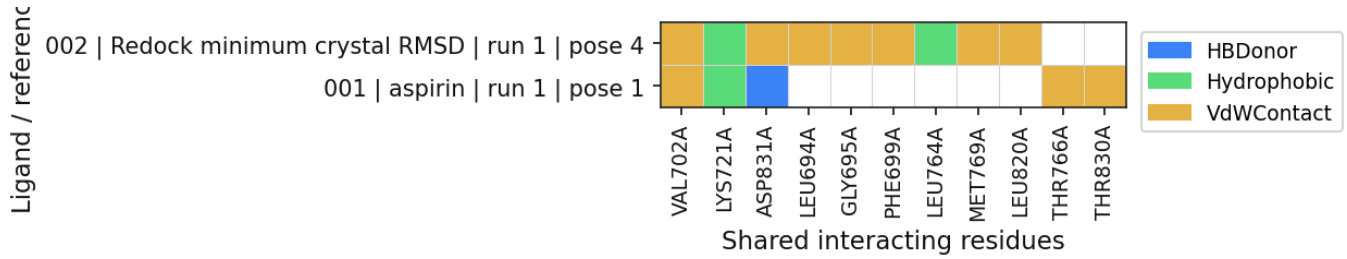
AQ4 redock

-7.30 kcal/mol

Lower score is more favorable in this scoring model; it is not a measured affinity.

Saved Section 4 plot; the red line marks the redocked co-crystal reference.

ProLIF interaction profile



Saved Section 4.1 profile: contact classes by residue for aspirin and the AQ4 redock.

Gemini interpretation - saved Section 4 output

Gemini compared the scores: aspirin is 1.339 kcal/mol less favorable than the AQ4 redock by Vina. This is a predicted score comparison.

With one run, variability and rank stability cannot be assessed. The score-table RMSD of 0.0 is a pose self-comparison, not crystal-pose validation.

Review AI comments against the saved files. Section 5 ZIP output was absent from the public copy.

Understand your results

Read a score, inspect a pose, and export the files you need.

AFFINITY

-7.32 kcal/mol

Vina's predicted score. More negative ranks better within this run.

RMSD

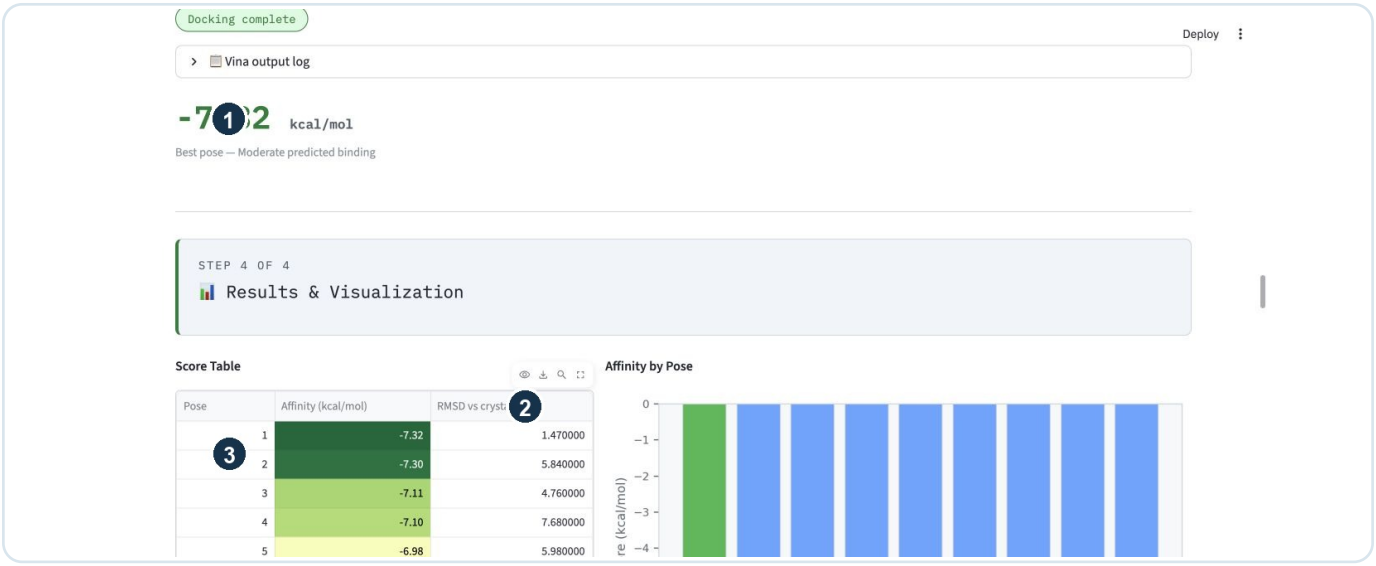
1.47 Å

Distance from the crystal reference for this example pose.

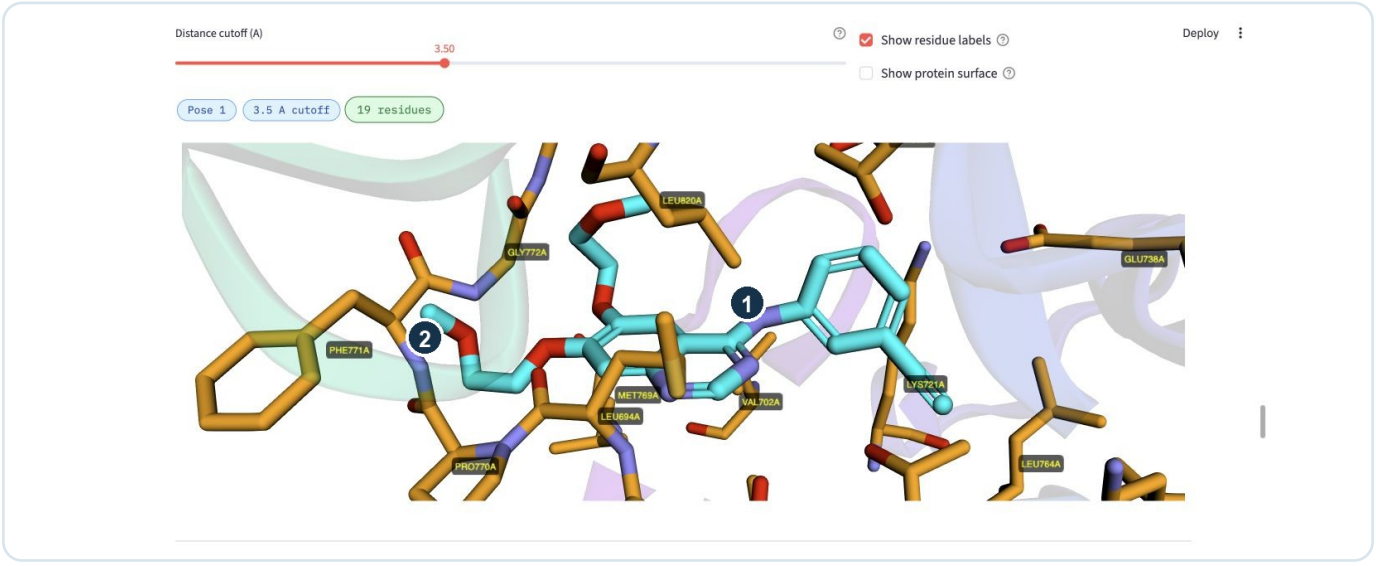
POSE

1 / 9

A predicted 3D position and orientation of erlotinib.



Online demonstration run: 1 score 2 RMSD column 3 pose rows



3D pocket view of pose 1: 1 cyan erlotinib 2 gold nearby residues

Download: choose a pose (.sdf or .pdb), all poses (.pdbqt), scores (.csv), or receptor (.pdb) from the result panel.
Scientific caution: a docking score does not prove experimental binding affinity.